

Homework 3

Deadline: Mar. 10, 2022

2.17(a)

2.17 (a) Calculate the final pressure of a sample of carbon dioxide that expands reversibly and adiabatically from 57.4 kPa and 1.0 L to a final volume of 2.0 L. Take $\gamma = 1.4$.

2.24(b)

2.24 (b) A sample consisting of 2.5 mol of gas molecules at 220 K and 200 kPa is compressed reversibly and adiabatically until the temperature reaches 255 K. Given that its molar constant-volume heat capacity is $27.6 \text{ J K}^{-1} \text{ mol}^{-1}$, calculate q , w , ΔU , ΔH , and the final pressure and volume.

Self-test 2.9

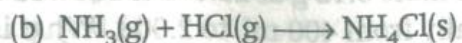
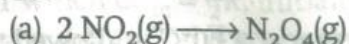
Estimate the standard enthalpy of formation of cyclohexane (liquid) at 400 K from the data in Table 2.5 at the end of book.

E2.30b

2.30 (b) The standard enthalpy of formation of phenol is $-165.0 \text{ kJ mol}^{-1}$. Calculate its standard enthalpy of combustion.

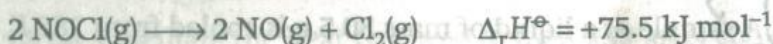
E2.39a

2.39 (a) Use standard enthalpies of formation to calculate the standard enthalpies of the following reactions:



E2.42b

2.42 (b) Calculate the standard enthalpy of formation of $\text{NOCl}(\text{g})$ from the enthalpy of formation of NO given in Table 2.5, together with the following information:



1. Write down the perfect gas law.
2. Explain the term “partial pressure”
3. Explain the following terms: “open system” “closed system” “isolated system” “state function” and “internal energy”

4. Explain the difference between “isothermal” and “adiabatic” processes
5. A sample consisting of 3.7 moles of ideal gas molecules with $C_{p,m} = 17.5 \text{ J K}^{-1} \text{ mol}^{-1}$ is initially at 173 kPa and 323 K. It then undergoes a reversible adiabatic expansion until its temperature is 247K.
 - 1) What is the final volume and pressure?
 - 2) What is heat and work for this process?
6. For each of the following transformations, determine whether q, w, ΔU , and ΔH of the system are greater than zero, less than zero, or equal to zero.
 - (1) Air is sealed in a container with rigid and then heated from room temperature to 90°C
 - (2) Gas in an adiabatic cylinder 5.0 L in volume is compressed by a constant external pressure of 4.0 atm to a final volume of 1.0 L.
7. Carbon dioxide (CO_2) and H_2O can be produced from the reaction between CH_4 and O_2 . Answer the following questions using data given below.
 - 1) Calculate the standard enthalpy of the reaction at 298K under 1 bar.
 - 2) Figure out the molar reaction enthalpy at the temperature of 500 K.

$$\Delta_f H^\ominus(\text{CO}, \text{g}) = -110.53 \text{ kJ mol}^{-1}$$

$$\Delta_f H^\ominus(\text{CO}_2, \text{g}) = -393.51 \text{ kJ mol}^{-1}$$

$$\Delta_f H^\ominus(\text{CH}_4, \text{g}) = -74.81 \text{ kJ mol}^{-1}$$

$$\Delta_f H^\ominus(\text{H}_2\text{O}, \text{l}) = -285.83 \text{ kJ mol}^{-1}$$

$$C_{p,m}(\text{CH}_4, \text{g}) = 35.31 \text{ J K}^{-1} \text{ mol}^{-1}$$

$$C_{p,m}(\text{CO}_2, \text{g}) = 37.11 \text{ J K}^{-1} \text{ mol}^{-1}$$

$$C_{p,m}(\text{O}_2, \text{g}) = 36.16 \text{ J K}^{-1} \text{ mol}^{-1}$$

$$C_{p,m}(\text{CO}, \text{g}) = 29.14 \text{ J K}^{-1} \text{ mol}^{-1}$$

$$C_{p,m}(\text{H}_2\text{O}, \text{l}) = 75.291 \text{ J K}^{-1} \text{ mol}^{-1}$$

$$C_{p,m}(\text{H}_2\text{O}, \text{g}) = 33.58 \text{ J K}^{-1} \text{ mol}^{-1}$$

$$\Delta_{\text{vap}} H^\ominus(\text{H}_2\text{O}) = 40.656 \text{ kJ mol}^{-1}$$

以下为选做内容

2.4 A 5.0 g block of solid carbon dioxide is allowed to evaporate in a vessel of volume 100 cm^3 maintained at 25°C . Calculate the work done when the system expands (a) isothermally against a pressure of 1.0 atm, and (b) isothermally and reversibly to the same volume as in (a).

P 2.5

2.5 A sample consisting of $1.0\text{ mol CaCO}_3(\text{s})$ was heated to 800°C , when it decomposed. The heating was carried out in a container fitted with a piston that was initially resting on the solid. Calculate the work done during complete decomposition at 1.0 atm. What work would be done if instead of having a piston the container was open to the atmosphere?

P 2.11

2.11 An object is cooled by the evaporation of liquid methane at its normal boiling point (112 K). What volume of gas at 1.00 atm

pressure must be formed from the liquid in order to remove 32.5 kJ of energy as heat from the object?

P2.30

2.30 Glucose and fructose are simple sugars with the molecular formula $\text{C}_6\text{H}_{12}\text{O}_6$. Sucrose, or table sugar, is a complex sugar with molecular formula $\text{C}_{12}\text{H}_{22}\text{O}_{11}$ that consists of a glucose unit covalently bound to a fructose unit (a water molecule is released as a result of the reaction between glucose and fructose to form sucrose). (a) Calculate the energy released as heat when a typical table sugar cube of mass 1.5 g is burned in air. (b) To what height could you climb on the energy a table sugar cube provides assuming 25 per cent of the energy is available for work? (c) The mass of a typical glucose tablet is 2.5 g . Calculate the energy released as heat when a glucose tablet is burned in air. (d) To what height could you climb on the energy a cube provides assuming that 25 per cent of the energy is available for work?

P 3.6

3.6† A cylindrical container of fixed total volume is divided into three sections, S_1 , S_2 , and S_3 . The sections S_1 and S_2 are separated by an adiabatic piston whereas S_2 and S_3 are separated by a diathermic (heat-conducting) piston. The pistons can slide along the walls of the cylinder without friction. Each section of the cylinder contains 1.00 mol of a perfect diatomic gas. Initially, the gas pressure in all three sections is 1.00 bar and the temperature is 298 K . The gas in S_1 is heated slowly until the temperature of the gas in S_3 reaches 348 K . Find the final temperature, pressure, and volume, as well as the change in internal energy for each section. Determine the total energy supplied to the gas in S_1 .